

1. Practical5b.c

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    pipe(fd);

    if(fork()==0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    if(fork()==0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

```

myshell> nano pac.c
myshell> gcc pac.c -o ls
myshell> ./ls
-rwxr-xr-x 1 varshini varshini 16040 Oct  5 08:24 ab
-rw-r--r-- 1 varshini varshini  221 Oct  5 16:15 c
-rw-r--r-- 1 varshini varshini    0 Aug 13 15:34 client_fi
-rw-r--r-- 1 varshini varshini  49 Oct  5 09:22 combined.
-rw-r--r-- 1 varshini varshini  692 Oct  5 10:09 fcfs.c
-rw-r--r-- 1 varshini varshini  220 Oct  5 16:16 fifo.c
-rw-r--r-- 1 varshini varshini  252 Oct  5 16:19 fifor.c
-rw-r--r-- 1 varshini varshini  291 Oct  5 16:20 fifow.c
-rw-r--r-- 1 varshini varshini   78 Aug  4 03:47 hello.c
-rw----- 1 varshini varshini  117 Jul 31 09:36 hello.c.s
-rw-r--r-- 1 varshini varshini   78 Aug  4 03:46 hello.c.s
-rwxr-xr-x 1 varshini varshini 22000 Sep 28 04:34 ipc
-rwxr-xr-x 1 varshini varshini 22000 Sep 28 04:33 ipc_proje

```

2. Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.

```

#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handler(int sig)
{
    if(sig==SIGINT)
        printf("\nSIGINT Received\n");

    if(sig==SIGTERM)
        printf("\nSIGTERM Received\n");

    if(sig==SIGUSR1)
        printf("\nSIGUSR1 Received\n");
}

int main()
{
    signal(SIGINT,handler);
    signal(SIGTERM,handler);
    signal(SIGUSR1,handler);

    while(1)
    {
        printf("Running...\n");
        sleep(3);
    }

    return 0;
}

```

```
myshell> nano pac6.c
myshell> gcc pac6.c -o sl
myshell> ./sl
Running...
Running...
Running...
Running...
Running...
```

Ex; kill -SIGUSR1 PID

Ex: kill -SIGTERM PID